RESEARCH ARTICLE | JUNE 09 2026

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AIP Advances 16, 065206 (2026)
https://doi.org/10.1063/5.0327634

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Cite as: AIP Advances 16, 065206 (2026); doi: 10.1063/5.0327634
Submitted: 12 February 2026 • Accepted: 16 May 2026 •
Published Online: 9 June 2026

Soomin Kim,[1,a)] Hanggyo Jung,[2,b)] Jong-Min Lee,[3,c)] Jong Yul Park,[3,d)] Junhyung Jeong,[3,e)] Dong-Min Kang,[3,f)] and Jongwook Jeon[4,g)]

AFFILIATIONS

[1] Department of Display Engineering, Sungkyunkwan University, Suwon 16419, South Korea
[2] Department of Semiconductor Convergence Engineering, Sungkyunkwan University, Suwon 16419, South Korea
[3] RF/Power Components Research Section, Electronics and Telecommunications Research Institute (ETRI), Daejeon, South Korea
[4] Department of Electrical and Computer Engineering, Sungkyunkwan University, Suwon 16419, South Korea

[a)] **Email:** mini100717@skku.edu
[b)] **Email:** gkdry@skku.edu
[c)] **Email:** leejongmin@etri.re.kr
[d)] **Email:** jongyulpark@etri.re.kr
[e)] **Email:** jjunh05@etri.re.kr
[f)] **Email:** kdm1597@etri.re.kr
[g)] **Author to whom correspondence should be addressed:** voix0707@skku.edu

ABSTRACT

This study first detailed simulation-based α-particle-induced effects in AlGaN/gallium nitride (GaN) high electron mobility transistors (HEMTs), with a specific focus on the correlation between gate/drain transient currents and trap behaviors. We quantitatively analyzed single-event effects (SEE) responses under varying gate/drain biases, ion strike locations, and trap types/densities to investigate the device's SEE sensitivity and charge collection mechanisms. The results show that radiation sensitivity depends heavily on these conditions, with the device being particularly vulnerable in the OFF-state. In particular, drain transient current was most sensitive to strikes at the gate–drain edge, while gate current peaked at the gate center. To mitigate these effects, field-plate architecture was introduced to effectively redistribute peak electric fields, thereby providing a robust design guideline for improving the radiation tolerance of GaN HEMTs. Crucially, acceptor traps at the AlGaN/GaN interface were identified as the governing factor. Trapped electrons distort the local electric field, inducing band bending and activating the gate leakage mechanism. These results provide a comprehensive analysis of alpha particle-induced effects in AlGaN/GaN HEMTs, focusing on gate leakage and trap dynamics, addressing a gap in prior research. These results offer essential insights and establish a theoretical foundation for radiation-hardened device design.

I. INTRODUCTION

Gallium nitride (GaN), a III–V group nitride semiconductor, has been receiving a lot of attention as a promising next-generation semiconductor material due to its wide bandgap of 3.4 eV and high electric field strength.[1–3] Among GaN-based devices, the AlGaN/GaN high electron mobility transistor (HEMT) achieves high-power and high-frequency operation with low power loss, thanks to the high mobility of the two-dimensional electron gas (2DEG) formed at the AlGaN/GaN heterojunction and a high critical electric field, thereby enabling a high breakdown voltage and superior power handling capability.[4–8]

Meanwhile, when devices operate in extreme environments, such as aerospace, device reliability degradation caused by radiation becomes a critical concern.[9] The International Technology Roadmap for Semiconductors (ITRS, 2015) also identified environmental radiation as a key reliability concern.[10] Radiation damage is broadly classified into two categories: (1) total ionizing dose (TID) and (2) single event effects (SEE).[11–13] While many studies have focused on the effects of TID on GaN-based transistors,[14–17] some research suggests that these devices may be relatively sensitive to SEE. SEE occurs when a high-energy particle traverses the device and deposits energy along its path within the semiconductor material. The large number of electron–hole pairs (EHPs) generated in this process either recombine or move via drift and diffusion, inducing a temporary transient current pulse. This transient current pulse, known as a single event transient (SET), can temporarily disrupt the normal operation of the device.[18–23] In addition, thermodynamic analyses have also been introduced to quantify the transient lattice heating accompanying particle strikes.[24]

SEE research on GaN HEMTs has primarily centered on heavy-ion irradiation. Abbate *et al.* irradiated commercial normally-off GaN power HEMT devices with ratings from 40 to 200 V, with various heavy ions and monitored changes in drain leakage currents in real-time.[25] Onoda *et al.* focused on elucidating the mechanism of enhanced charge collection in GaN HEMTs exposed to heavy ions, where significantly more charge is collected than predicted.[26]

While the SET degradation response to heavy-ion irradiation in HEMTs has been relatively systematically studied, reports on the effects of α-particles on AlGaN/GaN devices are scarce. However, α-particles represent a non-negligible radiation component, constituting ~9% of the particle flux in the Van Allen belts.[27] Furthermore, it has been reported that in terrestrial and near-terrestrial environments, electronic systems near large facilities such as nuclear fusion reactors can be exposed to a fluence of $\sim 10^{14}$ cm^{-2} per year for 2.45 and 14 MeV neutrons and α-particles, respectively.[28] This implies that radiation-tolerant design for GaN HEMT devices is essential and underscores the necessity for research into SEE induced by α-particles.

Direct experiments using radiation particles are time-consuming and costly, making comprehensive and repeatable parameter analysis difficult. Due to these constraints, technology computer-aided design (TCAD) simulation is widely used as a tool to investigate physical mechanisms and analyze radiation effects.[29,30] However, to realize GaN-based HEMT devices with higher radiation hardness, in-depth analysis surpassing the limitations of prior research is required.

1. Previous studies on radiation effects have primarily focused on analyzing drain transient currents.[31,32] However, there is a lack of detailed analysis regarding the specific mechanisms that induce gate leakage current during α-particle strikes despite its critical impact on the device's input stability and long-term reliability.
2. Furthermore, there has been no systematic and quantitative analysis of how charge trapping behaviors and single-event transient (SET) responses vary, depending on the location and type of various traps within the device that degrade performance.

Therefore, this study aims to comprehensively analyze the impact of α-particle strikes on AlGaN/GaN HEMTs, focusing on trap-related mechanisms governing gate leakage and SET responses utilizing TCAD simulations. To this end, we systematically vary and compare the effects of the following detailed conditions: (i) various operating regions by changing gate and drain bias conditions, (ii) the strike location of the α-particle, and (iii) a detailed classification of four types of traps existing within the device, with trap density as a variable for each to investigate the corresponding charge trapping behavior. In particular, by adopting a field-plate design, this study quantitatively evaluated the resulting enhancement in radiation-hardened performance. Through this approach, we aim to elucidate the mechanisms by which trap characteristics influence gate leakage current fluctuations and SET generation and offer design guidelines for radiation-hard GaN-HEMT devices.

The structure of this paper is as follows: Sec. II provides a detailed description of the AlGaN/GaN HEMT device modeling method, physical model parameters, and the α-particle modeling procedure. Subsequently, Sec. III discusses the simulation results of the SET response under the various conditions defined earlier. Finally, Sec. IV presents the conclusions. The findings of this study are expected to contribute to the space application and radiation-hardening design of GaN HEMT devices.

II. METHOD

To analyze the radiation effects on the AlGaN/GaN HEMT device, a two-dimensional (2D) device structure was constructed using Synopsys Sentaurus TCAD software. Various physical models were applied to the constructed model to reflect its electrical characteristics and radiation-related physical phenomena.

In this study, the radiation hardness of the device was evaluated based on the process flow presented in Fig. 1(a). First, the geometric structure of the device was defined according to measurement data, and the DC I–V calibration was performed by systematically adjusting key physical parameters, such as the gate metal work function, polarization charges, and trap density/energy levels. Second, the depth-dependent stopping profile of a 6 MeV α-particle within the AlGaN/GaN layer stack was extracted via TRIM Monte Carlo simulation. Third, based on the TRIM results, the Sentaurus Alpha Particle generation model was independently re-calibrated, after which transient response simulations were carried out under specific irradiation conditions including the particle's energy, strike location, time, and incidence angle. Since the DC I–V calibration focuses on validating the steady-state device physics, the second and third calibration steps were performed separately to more reliably reproduce the transient SET response induced by particle strikes.

A. Device structure and parameters

The HEMT device used in this study consists of an AlGaN/GaN heterostructure fabricated on a semi-insulating 4H–SiC substrate. The epitaxial structure is composed of a 2000 nm thick Fe-doped GaN buffer layer and an AlGaN barrier layer with an aluminum (Al) content of 25.5%. A 50 nm silicon nitride (SiN) passivation layer was deposited on the device surface to suppress surface states and enhance stability.

The key specifications of the HEMT include the gate length, unit gate width, source length, and drain length. Table I lists the detailed parameters used in the simulation. Low-resistance ohmic contacts for the source and drain were deposited in a multilayer

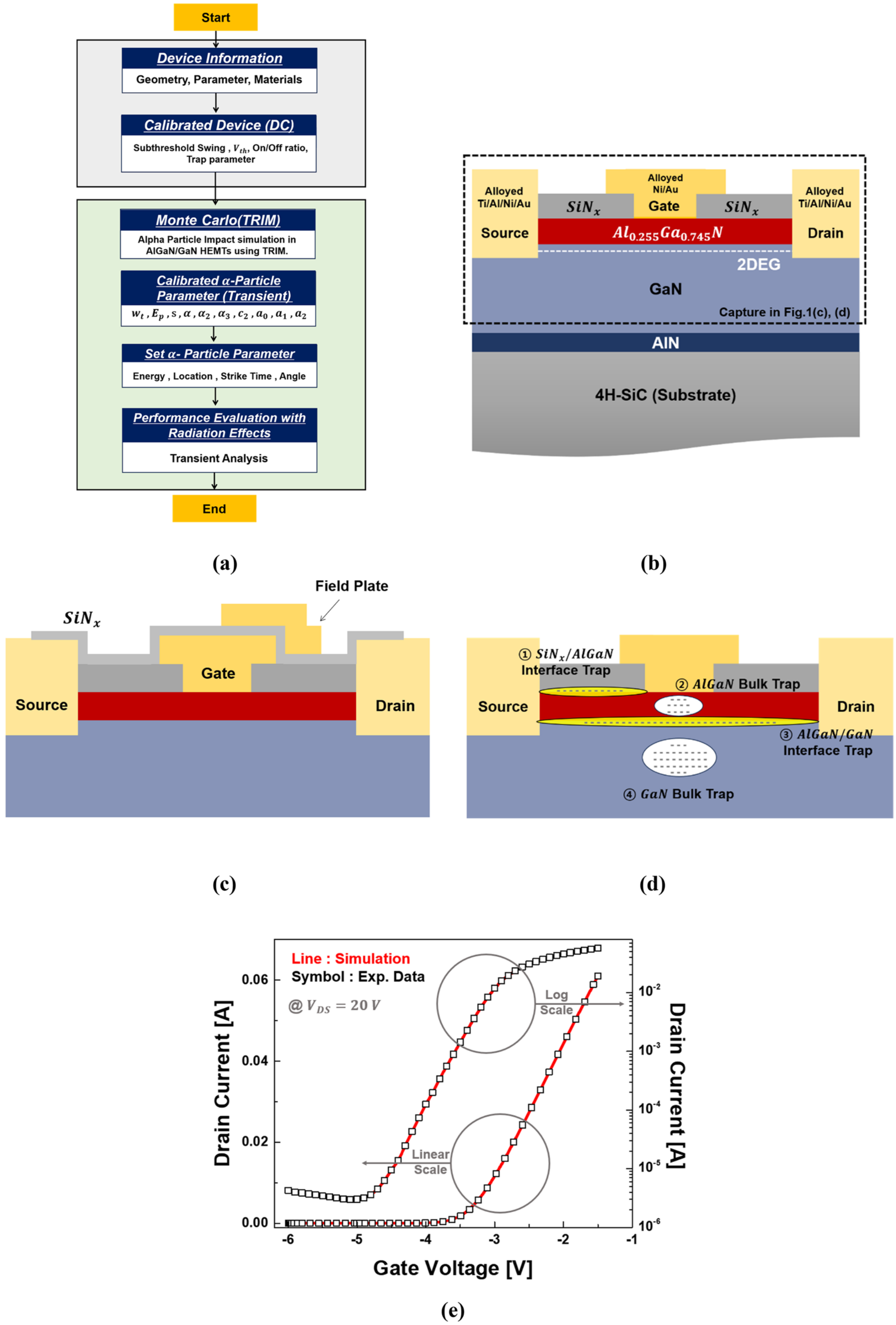

FIG. 1. Process flow and schematic illustrations of AlGaN/GaN HEMTs and calibration results. (a) Process flow for single event effect evaluation using TCAD simulation. (b) Device model of AlGaN/GaN HEMTs. (c) Device with a field plate. (d) Four types of traps considered. (e) Calibration results for V_g–I_d curves of AlGaN/GaN HEMTs.

TABLE I. Key device parameters used in this work.

Dimension parameter	Value (μm)
Source length (L_s)	23
Drain length (L_D)	15
Gate length (L_G)	0.15
AlGaN thickness	0.025
Field plate height	0.44
Field plate length	1.5

TABLE II. Parameter of material used in this work.

Parameter of material	GaN
Electron mobility (μ_n)	1500 (cm^2/V s)
Energy bandgap (E_g)	3.4 (eV)
Relative permittivity (ε)	9.5 (-)
Saturation velocity (V_{sat})	2.6 (10^7 cm/s)

structure of Ti/Al/Ni/Au (30/100/30/100 nm), and the T-shaped gate electrode is composed of Ni/Au (30/400 nm). The Ti/Al/Ni/Au multilayer ensures low contact resistance, while the T-shaped Ni/Au gate minimizes gate resistance and improves high-frequency performance. To deeply analyze the impact of electric field modulation on the SET mechanism, a field-plate-equipped HEMT was modeled. The field plate was constructed by depositing a metal layer on the SiN_x passivation, thereby extending the electrode over the gate-to-source and gate-to-drain access regions. The calibrated two-dimensional (2D) device cross section used for TCAD simulations is illustrated in Figs. 1(b) and 1(c).

B. Physical models and calibration

Because most of the charge transport and field modulation in HEMTs occur in the lateral plane, a 2D cross-sectional model was constructed and modeled in Synopsys' Sentaurus TCAD software. The 2D sheet charge model assumes a worst-case scenario by neglecting lateral diffusion, leading to more severe transient responses than 3D environments. Since SET is most critical during vertical ion incidence, a phenomenon effectively captured in 2D, this modeling approach offers sufficient validity and reliability for radiation tolerance analysis.

To ensure the model's physical validity and predictive accuracy, a precise calibration was performed based on actually measured DC I–V characteristics. The carrier transport models in TCAD utilize advanced physical models provided by Sentaurus software.

Carrier transport was modeled using the drift–diffusion (DD) framework combined with a high-field-saturation mobility model to account for velocity-saturation effects under strong electric fields. Furthermore, Fermi–Dirac carrier statistics were employed to ensure accuracy in high-carrier-density regions. To accurately simulate trap-assisted-recombination due to defect states within the device, concentration-dependent Shockley–Read–Hall (SRH) and Auger recombination models were introduced. An impact-ionization model was incorporated to describe carrier multiplication resulting from α-particle strikes, and the incident particle was set as an α-particle model. The work function of the gate contact and the spontaneous and piezoelectric polarization values of the AlGaN and GaN layers were systematically adjusted to minimize the error between the simulated and experimental results for the threshold voltage and maximum drain current. All physical models used in this simulation were implemented within the standard models provided by Sentaurus TCAD. Table II lists the detailed electrical material properties used in the simulation.

To precisely model the defect states that significantly influence the device's electrical characteristics, traps were defined at four key physical locations, as shown in Fig. 1(d).[33–36] This approach plays a decisive role in reproducing the experimentally observed current collapse and threshold shift. The density, energy level, and spatial distribution of each trap were optimized by using them as indicators to improve the agreement with experimental results. First, to implement the semi-insulating properties, fixed acceptor-like traps were set in the GaN buffer layer to control the off-state leakage current and breakdown voltage characteristics. Second, acceptor-like bulk traps were placed in the AlGaN layer to reflect defects arising during the epitaxial growth process, thereby modeling dynamic performance degradation such as on-resistance increase and maximum drain current reduction. Third, at the AlGaN/GaN heterojunction interface where the 2DEG channel is formed, acceptor-like interface traps with a broad energy distribution near the conduction band were defined to adjust the subthreshold swing (SS) and delay behavior in the subthreshold region. Prior studies have reported that acceptor-type interface traps dominate the turn-on transient response of AlGaN/GaN HEMTs,[37] and similarly, recent TCAD work has explicitly placed acceptor trap levels at this interface to reproduce experimental device characteristics.[38] Some irradiation studies have reported that AlGaN/GaN heterojunction interface-related trap states can be generated or enhanced under radiation-induced degradation conditions.[39,40] Therefore, the high-density interface-trap condition adopted in this work is not intended to represent a typical interface state but was introduced as a conservative stress-test condition to broadly account for possible trap-induced effects, including those associated with cumulative total-ionizing-dose (TID) degradation, emulate an extreme interface-trapping scenario that may occur after irradiation, and evaluate the sensitivity of the SET response to acceptor-type interface traps. Finally, at the SiN/AlGaN interface, donor-like traps were set to simulate surface states that can arise from the passivation process, thereby controlling the surface electric field and 2DEG density.

The dynamic behavior of each trap was modeled according to the Shockley–Read–Hall (SRH) recombination theory,[41] and the detailed parameters used in the simulation are summarized in Table III. As can be seen in Fig. 1(e), the transfer characteristics calculated by applying this trap model showed a high degree of agreement when compared with the actual measured values.

C. α-particle irradiation and charge deposition model

To elucidate the physical mechanism underlying α-particle-induced degradation in AlGaN/GaN HEMTs, we first quantified the interaction between the incident particles and the device stack. For

TABLE III. Trap setting parameter used in this work.

	Type	Energy level (eV)	Reference concentration (cm^{-2})	Concentration variation (cm^{-2})
SiN_x/AlGaN interface trap	Donor-like	$E_C - 0.5$	1×10^{12}	1×10^{12}–1×10^{13}
AlGaN bulk trap	Acceptor-like	$E_C - 0.5$	1×10^{12}	1×10^{18}–1×10^{19}
AlGaN/GaN interface trap	Acceptor-like	$E_C - 0.8$	1.53×10^{13}	1.53×10^{13}–1×10^{14}
GaN bulk trap	Acceptor-like	$E_C - 0.6$	1×10^{18}	1×10^{18}–1×10^{19}

this purpose, the Monte Carlo simulation program TRIM (Transport of Ions in Matter)[42–45] was utilized to calculate the depth-dependent energy loss profile that occurs when α-particles penetrate the multilayer structure of the device. This simulation separately calculates the electronic stopping and nuclear stopping for the incident α-particles and their induced recoils, which correspond to linear energy transfer (LET) and non-ionizing energy loss (NIEL), respectively.

Figure 2(a) illustrates the results of a Monte Carlo simulation showing the energy loss of a 6 MeV α-particle as it strikes the device stack. Simulation results revealed that a 6 MeV α-particle interacting with the device lattice loses >99.7% of its energy through ionization, demonstrating that electronic stopping dominates the energy loss process. Physically, this corresponds to the process where the α-particle loses energy via inelastic Coulomb interactions with the orbital electrons of GaN atoms, generating high-density electron–hole pairs (EHPs) along its trajectory.[46] According to the Bethe–Bloch theory, as the particle's velocity decreases, its rate of energy loss per unit length (LET) increases, forming a Bragg peak that reaches its maximum just before the particle comes to rest.[47–49]

In this study, the Bragg peak was obtained through Monte Carlo simulations, assuming the 6 MeV α-particle is incident perpendicularly on the device surface. Because the spatial profile of the ionization track critically determines the transient response,[50–52] an accurate Bragg-curve representation was required for subsequent TCAD simulations. The characteristic model for an α-particle incident on Silicon, provided by TCAD as Eq. (1), was re-calibrated to fit the AlGaN/GaN HEMT model for the α-particle charge deposition established in this study,

$$G(u,v,w,t) = \frac{a}{\sqrt{2\pi}\cdot s}\exp\left[-\frac{1}{2}\left(\frac{t-t_m}{s}\right)^2 - \frac{1}{2}\left(\frac{v^2+\ w^2}{{w_t}^2}\right)\right] \times\left[c_1 e^{\alpha u} + c_2\exp\left(-\frac{1}{2}\left(\frac{u-\alpha_1}{\alpha_2}\right)^2\right)\right]. \tag{1}$$

The charge generation rate $G(u, v, w, t)$ induced by ion absorption in the Sentaurus device, given by Eq. (1), provides a spatiotemporal representation of how an α-particle generates electron–hole pairs (EHPs) as it passes through the device.[53–55] The excess car-

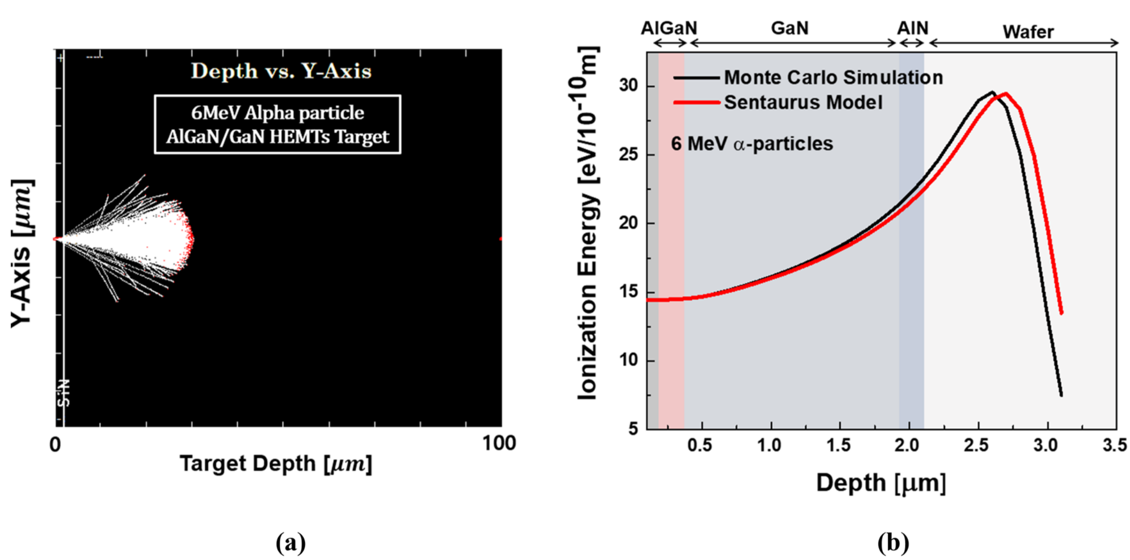

FIG. 2. Monte Carlo and TCAD calibrated results for radiation simulation: (a) α-particle impact simulation in AlGaN/GaN HEMTs using Monte Carlo simulation TRIM and (b) Bragg ionization curve of a 6 MeV α-particle in AlGaN/GaN HEMTs.

riers thus injected by the SEE are transported according to the self-consistent solution of the Poisson and drift–diffusion equations, and the resulting current waveform at the source/drain electrodes is computed as the SET current pulse. This model calculates the charge generation rate by combining the ion trajectory coordinates (l, w) based on cylindrical symmetry with a temporal Gaussian distribution (t_m, s). In particular, a scaling factor a is utilized to normalize the spatiotemporal integral of the generation rate to match the total generated charge, E/E_p. In this study, the conventional silicon-based model was re-calibrated for the AlGaN/GaN HEMT structure. In particular, the depth-dependent stopping-power profile and Bragg peak of a 6 MeV α-particle within the AlGaN/GaN HEMT structure were extracted via TRIM Monte Carlo simulation, and the parameters of the Sentaurus Alpha Particle model were re-calibrated against these results. This calibration approach is consistent with the prior TCAD study by Pocaterra and Ciappa, in which the Si-based Alpha Particle model was re-calibrated against Monte Carlo results for SiC Schottky power devices.[56] Its consistency was verified through a comparison of the stopping power curves with TRIM simulation results, as shown in Fig. 2(b). The calibrated parameters related to the α-particle are presented in Table IV.

TABLE IV. α-particle model parameters for TCAD simulation in this work.

Parameter	GaN value	SiC value
s	2×10^{-12} (s)	2×10^{-12} (s)
W_t	8×10^{-6} (cm)	1×10^{-5} (cm)
c_2	1.736	1.512
α	386.39 (cm^{-1})	275.0 (cm^{-1})
α_2	2.33×10^{-4} (cm)	2.003×10^{-4} (cm)
α_3	1.24×10^{-4} (cm)	1.15×10^{-4} (cm)
E_p	8.9 (eV)	7.8 (eV)
a_0	-1.279×10^{-4} (cm)	-1.135×10^{-4} (cm)
a_1	1.4681×10^{-10} (cm/eV)	1.7341×10^{-10} (cm/eV)
a_2	2.0149×10^{-17} (cm/eV^2)	2.9073×10^{-17} (cm/eV^2)

III. RESULTS AND DISCUSSION

A. Bias dependence of single-event transient (SET) response

When a high-energy α-particle strikes a sensitive region of a GaN HEMT, numerous electron–hole pairs (EHPs) are generated along its trajectory. These generated charge carriers are separated by the internal electric field of the device and are initially collected at their respective electrodes according to drift and diffusion mechanisms.[57]

In addition to the electric-field-driven drift–diffusion (DD) process, a secondary back-channel charge-amplification mechanism emerges under negative gate bias, as shown in Fig. 3. The holes generated by the particle strike move toward and accumulate in the buffer layer region beneath the gate electrode. The accumulation of these holes under the gate effectively lowers the potential barrier between the source and the channel.[58]

Once the potential barrier is lowered, electrons from the source electrode are easily injected over the barrier and into the channel. As a result, a continuous flow of electrons is established from the source to the drain. This current adds to the charge collected by the initial DD mechanism, thereby significantly increasing the total collected charge. Thus, in GaN HEMTs, the SEE is characterized by an amplification of the final collected charge due to the superposition of the drift–diffusion mechanism and the back-channel effect.

Therefore, the gate voltage directly influences the SET response by changing the device's operating point. Therefore, in this study, simulations were performed under two representative conditions: the On-state ($V_g = -2$ V) and the Off-state ($V_g = -4$ V).

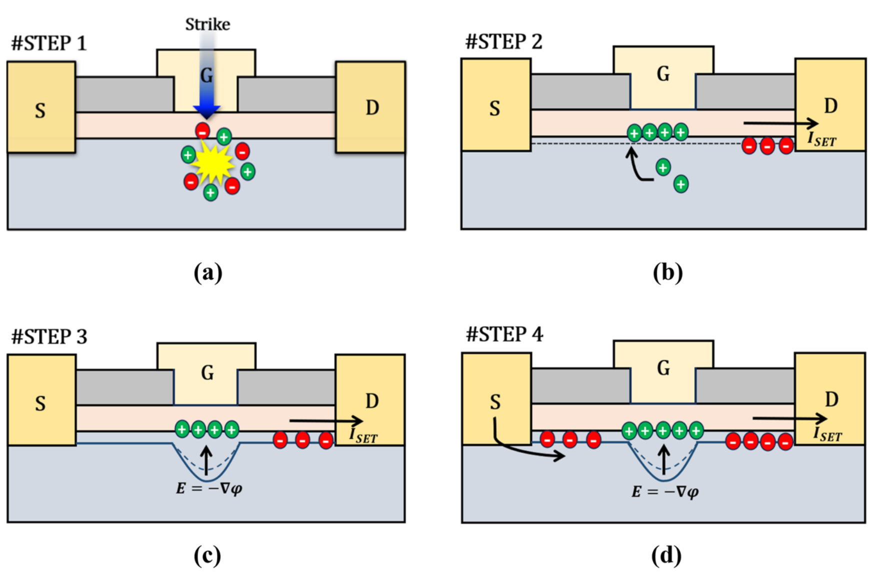

FIG. 3. Back channel effect. (a) Electron–hole pair (EHP) generation due to particle strike. (b) Initial charge collection by drift–diffusion (DD) and hole migration to the buffer layer. (c) Hole accumulation in the buffer layer beneath the gate, lowering the source-channel potential barrier. (d) Electron injection from the source over the lowered potential barrier, amplifying the total collected charge.

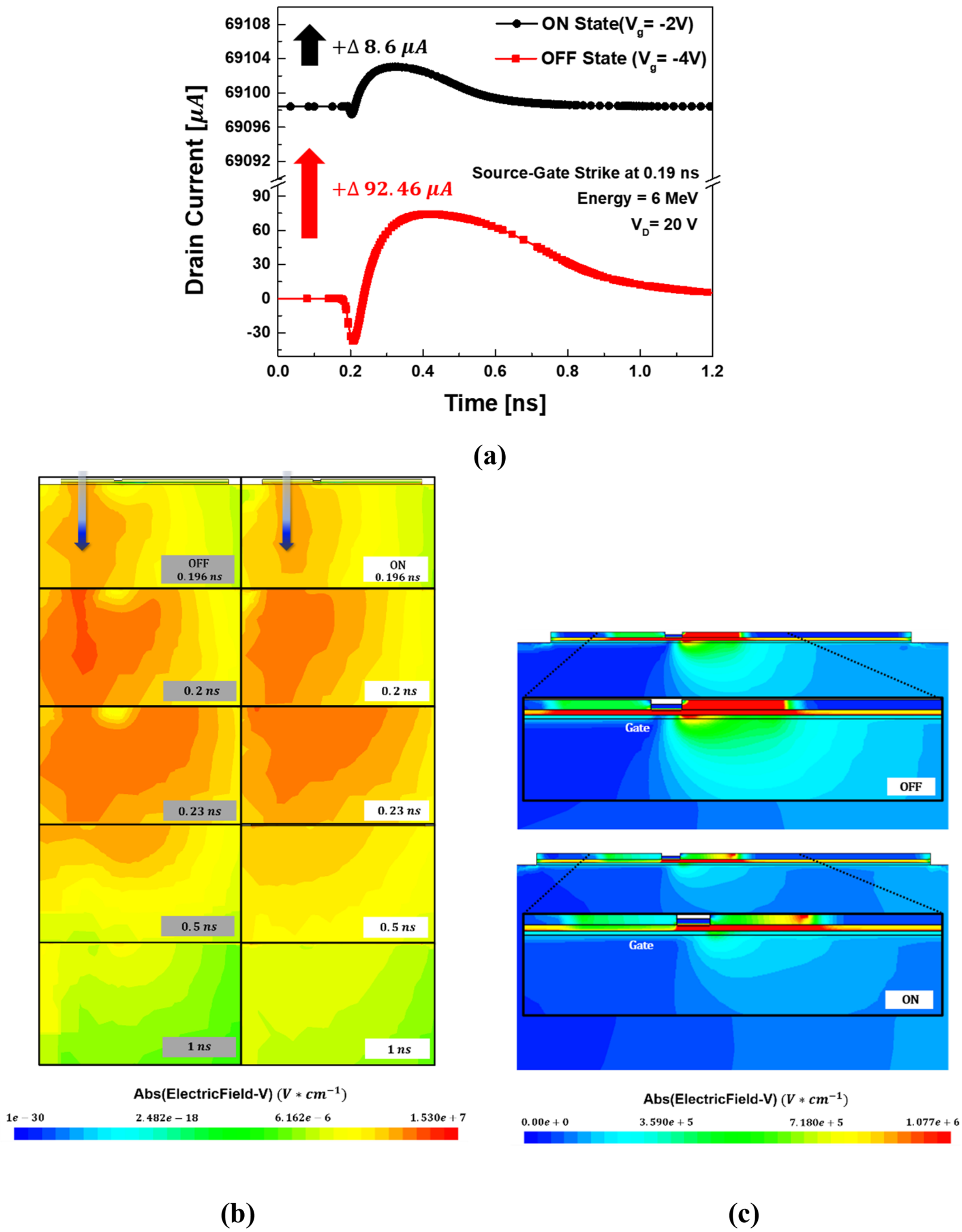

FIG. 4. Gate bias dependence of SET characteristics. (a) Transient drain current (I_d–time) characteristics of the AlGaN/GaN HEMT under On-state (V_g = −2 V) and Off-state (V_g = −4 V) bias condition. (b) Spatial distribution of electron current density immediately after α-particle impact. (c) Comparison of electric field distribution for On (V_g = −2 V) and Off (V_g = −4 V) bias conditions.

Figure 4(a) shows a comparison of the single event effect (SEE)-induced transient drain current in two operating modes, the On-state ($V_g = -2$ V) and the Off-state ($V_g = -4$ V), when a particle is vertically incident at a point between the source and the gate. In the Off-state, the drain current, which was 0.016 μA before the particle strike, rose to 92.48 μA after the strike. In the On-state, the current increased from 69095.1 μA pre-strike to 69103.7 μA post-strike. Accordingly, the Off-state current increased from 0.016 to 92.48 μA ($\approx 5.7 \times 10^3$ times), whereas the On-state increase was only 8.6 μA.

By integrating the transient current pulse, we can obtain the collected charge, Q_c, at the drain electrode. Q_c serves as a key metric for quantitatively analyzing the SET phenomenon in the time domain and indicates the radiation sensitivity of a specific node. The equation to calculate the charge collected solely due to the SET phenomenon, after removing the current flowing due to the initial bias, is given by the following equation:

$$Q_C = \int_{t_1}^{t_2} I(t)\ dt - \int_{t_1}^{t_2} I_{bias}\ dt. \qquad (2)$$

Calculating the collected charge for each case using the above-mentioned equation yields $Q_c \approx 6.60$ fC for the Off-state and 1.33 fC for the On-state. The collected charge in the Off-state is ~5 times greater than that in the On-state, confirming that the impact of SET is more pronounced in the Off-state than in the On-state.

Figure 4(b) illustrates the physical origin of this bias-dependent behavior by showing the spatiotemporal evolution of the electron current density J_n. As can be seen in the figure, when an α-particle strikes, high-density electron–hole pairs are generated along its trajectory. These charges are rapidly separated by the internal electric field and collected at the drain via drift–diffusion. Following the moment of impact (0.195 ns), the area and intensity of the region with high electron concentration inside the device temporarily expand and then rapidly contract as the charge is collected and recombines. The current pulse seen in Fig. 4(a) appears during this period, confirming that the temporal and spatial variation of the electron current density distribution is observed as the SET waveform.

Figure 4(c) shows the difference in the electric field distribution between the two operating points. In the Off-state, the negative voltage applied to the gate depletes the 2DEG channel, resulting in a very low electron concentration under the gate. The electron concentration in this region does not increase significantly until the charge is collected. The negative gate bias depletes the 2DEG channel, thereby blocking direct charge flow to the drain. Consequently, the charge is collected by the drain through a leakage path beneath the 2DEG channel, which has a more severe impact on the device. Conversely, when the device is in the On-state, the 2DEG channel is active, allowing the charge to be collected directly by the drain through the channel. This results in a relatively minor transient current phenomenon.

In summary, the observation of the electron current density and electric field distributions demonstrates that the gate bias is the key factor governing the SET response.

In this paper, the influence of the drain voltage on the single event transient (SET) in GaN HEMT devices was also analyzed. A transient response simulation was performed for the case of a particle vertically incident on the drain electrode. For this scenario, the gate voltage was held constant in the off-state ($V_g = -4$ V), while the drain voltage was varied from 10 to 27 V.

Figure 5(a) shows the transient drain current (I_D) waveforms corresponding to different drain voltages. A clear trend is observed

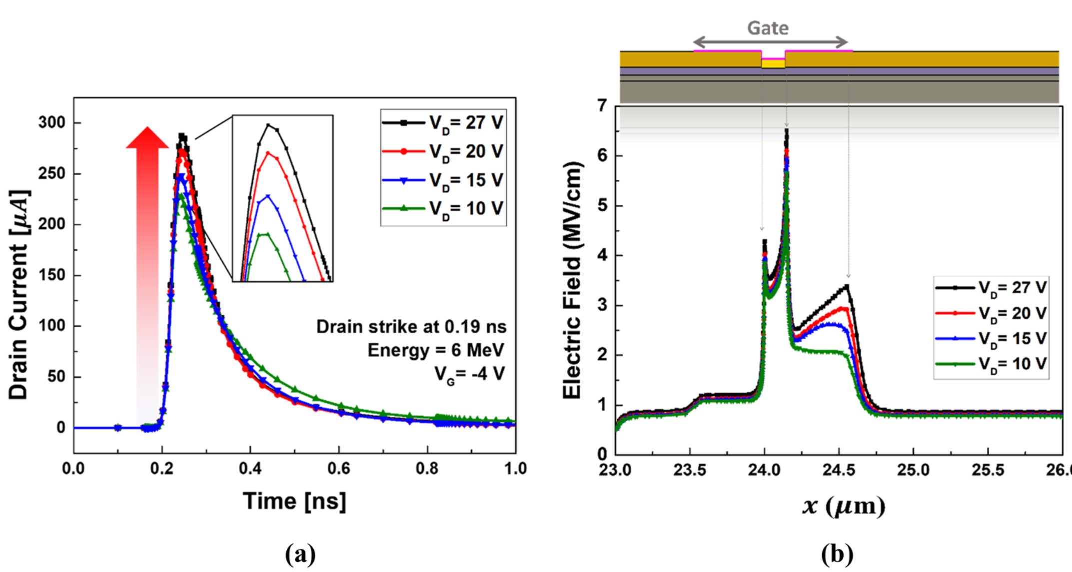

FIG. 5. Drain bias dependence of SET characteristics (V_d = 27, 20, 15, 10 V): (a) Transient drain current (I_d–time) responses to contact materials and (b) electric field distribution profiles corresponding to each drain bias condition.

where the peak value of the transient drain current increases as V_d increases. As V_d was raised to 10, 15, 20, and 27 V, the peak current values sequentially increased to 226, 247, 272, 295 μA, respectively. This increase in the transient current is directly related to the distribution of the lateral electric field within the device.

Figure 5(b) shows that increasing V_d intensifies the lateral electric field (E_x), particularly near the gate–drain edge, thereby enhancing carrier drift and charge-collection efficiency. A higher drain voltage leads to significantly stronger peak electric field intensity across the entire channel, particularly in the gate–drain edge region. This can be explained through the charge transport mechanism. The drift component of the electron current is described by the following equation:[59]

$$J_n = qn\mu_n E_x. \tag{3}$$

Here, E_x represents the lateral electric field. When the drain voltage increases, the E_x component becomes stronger. This exerts a greater drift force on the generated charge carriers, accelerating their movement toward the drain and thereby improving the charge collection efficiency. Consequently, a stronger electric field collects a larger amount of charge at the drain in a shorter amount of time. Consequently, stronger lateral fields yield larger and

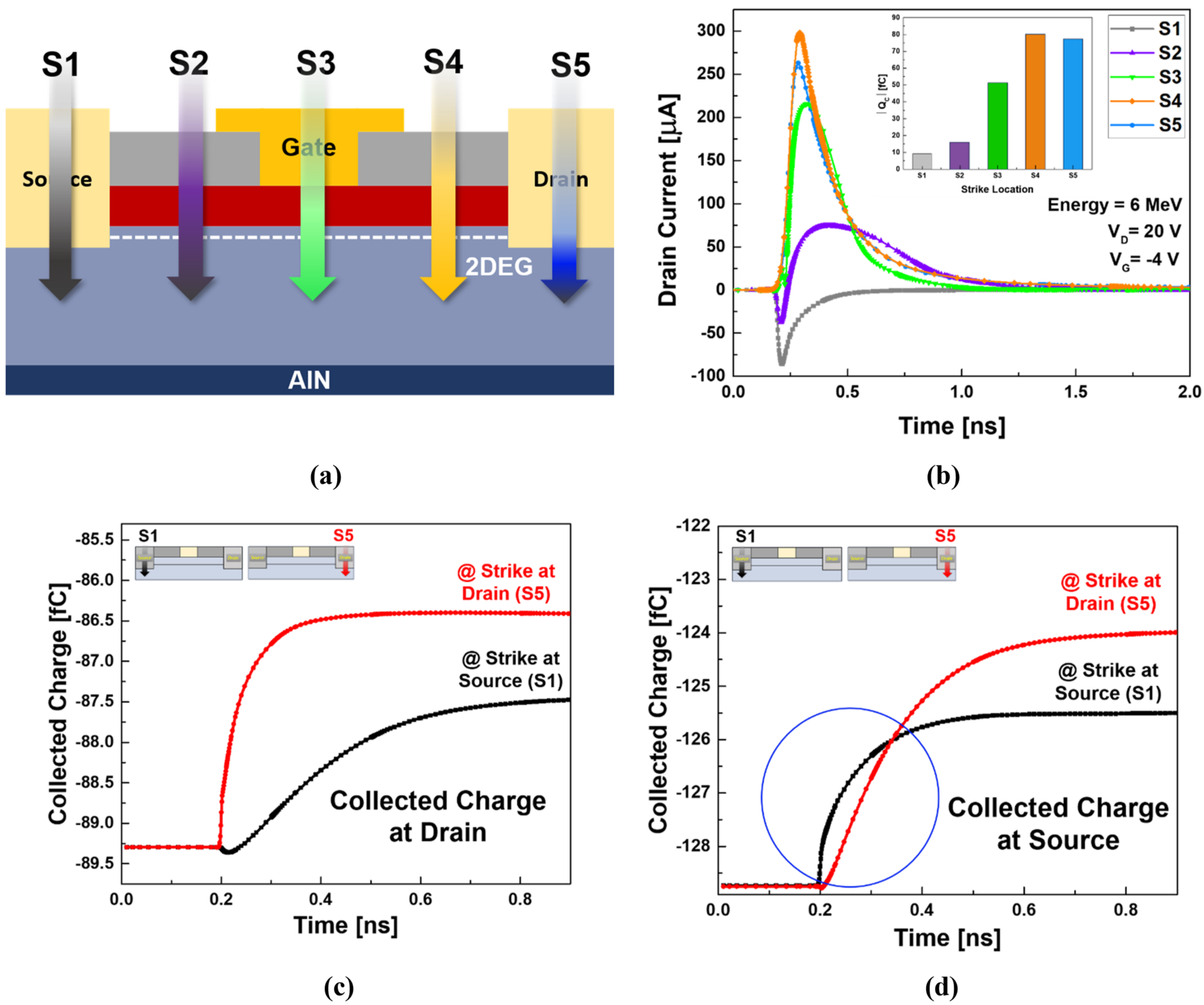

FIG. 6. Analysis of SET characteristics at five different particle strike locations: (a) Schematic illustration of the five incident positions (S1–S5). (b) Transient drain current (I_d–time) responses for each strike location. (c) Collected charge at the drain electrode under α-particle incidence. (d) Collected charge at the source electrode under the same strike conditions.

faster transient current peaks, consistent with the trend shown in Fig. 5(a).

B. Spatial sensitivity to particle strike location

The preceding simulation results were conducted based on conditions where the particle was incident between the source-gate region and at the center of the drain electrode. Since energetic particles may strike different regions of the device in realistic environments, this section investigates the spatial dependence of SET sensitivity.

Five strike locations were defined (S1: source electrode, S2: source-gate gap, S3: gate electrode, S4: gate-drain gap, S5: drain electrode) to systematically evaluate the location-dependent SET response, as shown in Fig. 6(a). These data are intended to identify the sensitive regions (hot spots) within the device for α-particles.

Figure 6(b) shows the transient drain current waveform and the total collected charge for each location. Both results indicate that the most potent SET response occurs when the strike is at the S4 (gate–drain edge) location. This high sensitivity at S4 originates from the strong internal electric field in that region, as was confirmed in the drain bias analysis in Sec. III A [see Fig. 5(b)]. The S4, or gate–drain edge, region is the point within the device where the electric field is most strongly concentrated. The transient current can be expressed as $I \approx dQ/dt$. At S4, carrier recombination is strongly suppressed and the charge transit time is minimized, leading to the steepest current rise and largest peak. This results in the steepest

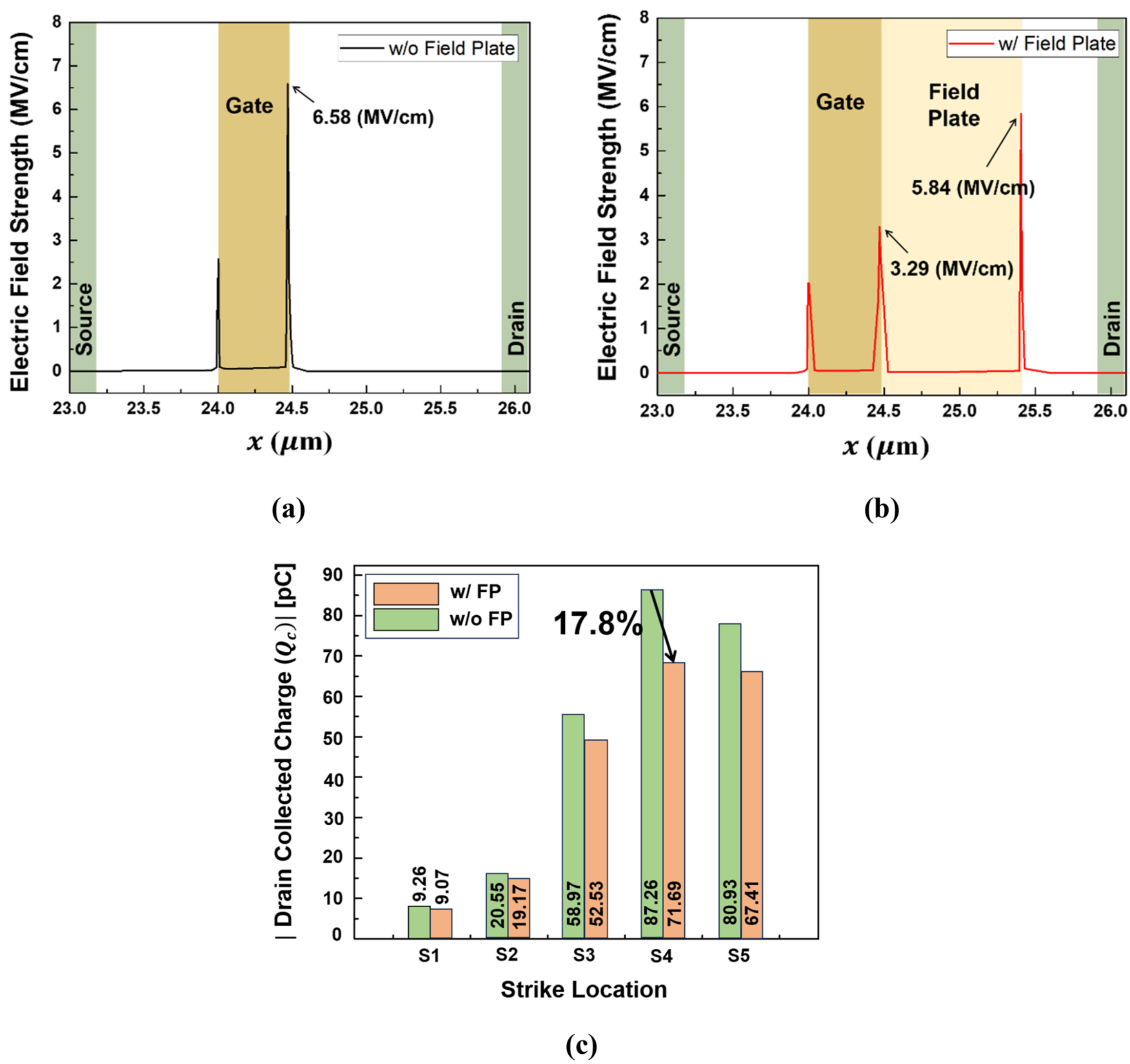

FIG. 7. Electric field distribution and collected charge at the drain electrode with and without a field plate: (a) Electric field intensity without a field plate. (b) Electric field intensity with a field plate (c) Comparison of collected charge at the drain electrode with and without a field plate.

initial rising slope of the current waveform and generates the largest peak value. In conclusion, it can be confirmed that the gate–drain edge region, where the internal potential difference is greatest and the electric field is concentrated, is the most sensitive area to the SET phenomenon.

A polarity reversal in the drain transient current is observed between S1 and S5 strikes. When a particle strikes S5, the generated electrons are immediately collected by the drain, resulting in a positive (+) current pulse. In contrast, when a particle strikes S1, the drain current temporarily takes on a negative (−) magnitude. This is because the initial collection path of the generated charge differs, depending on the strike location, and can be clearly explained through the charge collected at each electrode, as shown in Figs. 6(c) and 6(d). [In this simulation, current flowing into an electrode is defined as positive (+).]

First, in the case of a strike at S5 (drain), the electrons from the generated electron–hole pairs are immediately collected by the drain electrode. As indicated from the red curve shown in Fig. 6(c), the charge collected at the drain increases sharply immediately after the strike, which signifies a positive (+) current of electrons flowing into the drain.

On the other hand, the case of a strike at S1 (source) is different. The electrons generated by the strike are first collected rapidly by the nearest electrode, the source. The black curve shown in Fig. 6(d) indicates that a vast amount of charge is instantaneously collected at the source electrode upon an S1 strike. To preserve charge neutrality, the rapid influx of electrons at the source induces a transient displacement current of opposite polarity at the drain. This is why a negative (−) current pulse is initially observed at the drain. Subsequently, as some of the electrons generated near the source travel to the drain through the 2DEG channel, charge collection begins belatedly at the drain, as indicated by the black curve shown in Fig. 6(c), and the drain current switches back to a positive (+) value. Thus, the strike location acts as a key factor that determines the primary collection path of the generated charge, thereby changing the initial polarity and waveform of the observed transient current.

As established in the previous analysis, SET sensitivity is closely correlated with the internal electric field distribution, particularly the electric field concentration in the gate–drain region (S4). Therefore, in this section, a field plate (FP) structure was implemented to mitigate such electric field crowding as well as enhance radiation hardness and its effectiveness was systematically analyzed. Figure 7 illustrates the electric field distribution within the channel and the variation in collected drain charge for each strike location, both with and without the FP.

The calculated electric field intensity showed that the conventional structure (w/o Field Plate) in Fig. 7(a) exhibits a peak electric field of ~6.58 MV/cm at the gate–drain edge, confirming this region as the most vulnerable point for transient current generation upon radiation incidence. In contrast, for the structure featuring an FP (w/Field Plate) shown in Fig. 7(b), the peak electric field at the gate edge is reduced by ~2.3 times. Furthermore, the secondary peak electric field formed at the edge of the field plate is ~11% lower than the peak electric field of the conventional structure.

This field mitigation effect directly leads to a reduction in the charge collected via SET. The graph in Fig. 7(c) demonstrates that the collected charge decreases across all five strike locations (S1–S5) upon the application of the FP. Notably, at the S4 (gate–drain) position, previously identified as the most sensitive region, the collected charge was reduced by ~17.8%. This reduction is attributed to the mitigated electric field, which suppresses the separation and acceleration of the generated electron–hole pairs

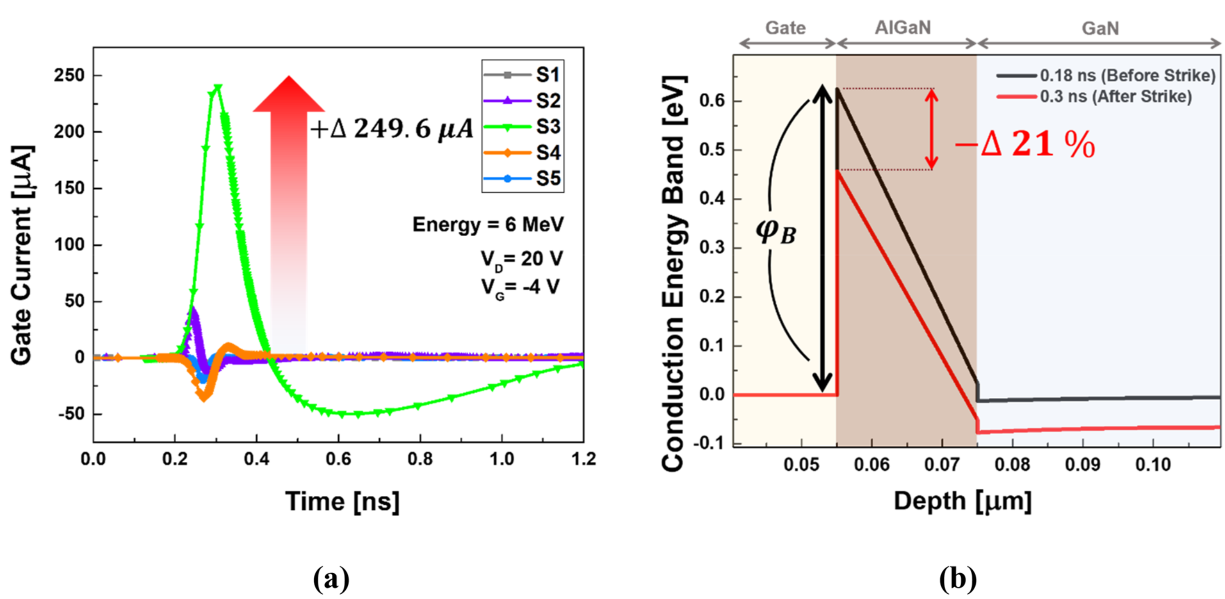

FIG. 8. Transient gate current mechanisms under particle strikes at five different locations: (a) Time-resolved gate current (I_g–time) responses for each incident position. (b) Conduction band energy profiles before and after particle incidence at the S3 (gate) location.

(EHPs) while increasing the recombination probability, thereby decreasing the total amount of charge collected at the drain. These results indicate that the field-plate structure is an effective solution for enhancing the SET immunity of the device against radiation.

In this study, we analyzed not only the drain transient current but also the impact of SET on the gate leakage current, a critical factor that governs device reliability. The gate of this device is formed as a Schottky contact on the AlGaN barrier layer. This metal–semiconductor interface is a highly sensitive region that directly controls the 2DEG channel and is a key point in determining the device's electrical isolation.[60–62] If a high-energy particle directly impacts the Schottky junction, severe transient or even permanent damage may occur. For this reason, investigating the effect of SET on gate leakage current is essential for evaluating the device's radiation hardness.

Figure 8(a) shows a comparison of the gate transient current according to the strike location (S1–S5). When an α-particle was vertically incident on S3 (the center of the gate electrode), the gate leakage current was observed to be the largest at 249.6 μA. This pattern differs from the previous result where the S4 region was most sensitive to the drain transient current.

Notably, the peak gate leakage current for the S3 strike (249.6 μA) is ~12.8% larger than the peak drain transient current

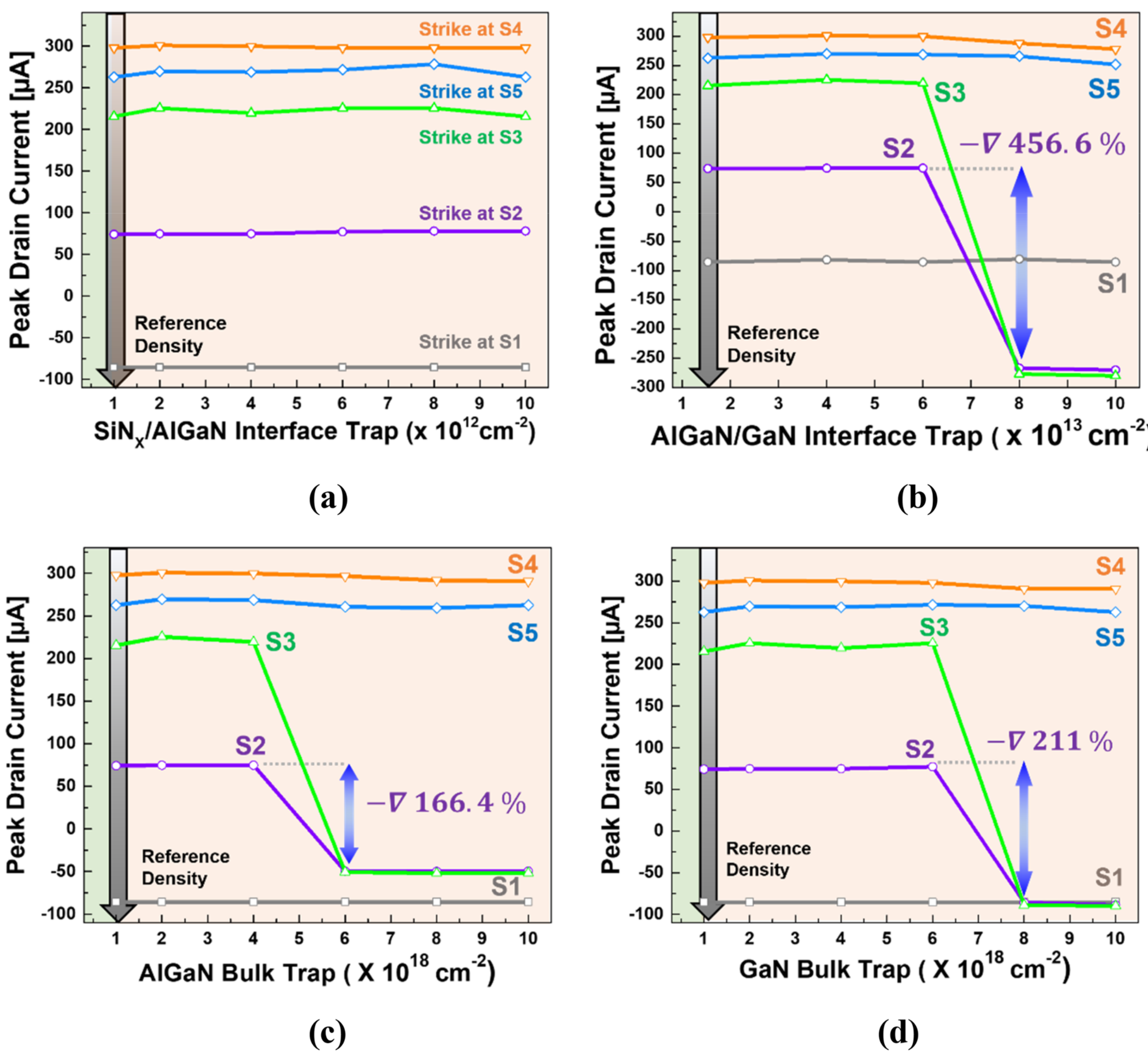

FIG. 9. Transient drain current peak analysis under increased trap densities at five different particle strike locations: (a) Variation with SiN_x/AlGaN interface traps density. (b) Variation with AlGaN/GaN interface trap density. (c) Variation with AlGaN bulk trap density. (d) Variation with GaN bulk trap density.

(219 μA) under the same conditions. This suggests that under certain circumstances, the SEE phenomenon can induce a more severe transient event in the gate electrode than in the drain.

The sharp increase in gate leakage is attributed to the localized damage track formed as the α-particle penetrates the Schottky barrier. In the case of an S3 strike, the particle directly and vertically penetrates the Schottky junction region between the gate metal and the AlGaN semiconductor, causing a localized collapse of the barrier in that area.

Figure 8(b) shows the conduction energy band profile observed at a vertical cross section (x = 0–0.11 μm) of the S3 location. The Schottky barrier height, which was 0.62 eV before the particle irradiation (0.18 ns), decreased by ~21% to 0.49 eV after the strike (t = 0.3 ns). This reduction in barrier height is due to defects induced by the particle strike within the bandgap and near the metal–AlGaN interface. Radiation-induced defects create electron tunneling paths via the trap-assisted tunneling (TAT) mechanism, substantially enhancing gate leakage. This trend is consistent with experimental reports of 10–16× increases in gate leakage due to radiation-induced Schottky barrier degradation in AlGaN/GaN HEMTs.[63]

C. Influence of traps on SET-induced gate leakage

Despite their excellent power and high-frequency characteristics, GaN HEMTs suffer from reliability issues due to defects (traps) within the device. As shown in Fig. 1(c), these traps exist in four distinct regions and alter the electron concentration of the 2DEG channel and the internal electric field distribution by capturing or emitting charges. Such charge-induced field variations lead to phenomena including current collapse, enhanced gate leakage, and threshold voltage instability.[64–68] In this section, we conduct an in-depth analysis of the impact of these four types of traps on the transient response characteristics from a single event transient (SET) perspective.

To isolate the influence of each trap type on transient behavior, the peak transient drain current was observed while increasing the trap density for five different strike locations, as shown in Fig. 9.

As shown in Fig. 9(a), even when the density of the SiN_x/AlGaN interface traps was significantly increased, the transient drain current value remained almost unchanged at all strike locations. This indicates that the capture and emission time constants of SiN_x/AlGaN interface traps are too long to influence SETs occurring on a picosecond–nanosecond scale. Therefore, the influence of SiN_x/AlGaN interface traps on the transient response is very limited.

In contrast, as can be seen in Figs. 9(b)–9(d), for the acceptor-type traps (AlGaN/GaN interface, AlGaN bulk, and GaN bulk), a distinct phenomenon was observed where the peak transient drain current decreased significantly for strikes near the gate (S2, S3) once the trap density exceeded a certain threshold value. High-mobility electrons generated by the particle strike are efficiently trapped by acceptor states near the gate region. The trapped electrons form localized negative charges that distort the band profile, depleting the 2DEG beneath the gate and suppressing drain current. This phenomenon is most prominent in the areas around the gate (S2, S3), where the electric field is concentrated.

In conclusion, the device's transient response characteristics are insensitive to SiN_x/AlGaN interface traps but react very sensitively to the density of acceptor traps located near the gate. This suggests that controlling acceptor-like defects is a key design factor for ensuring device reliability in radiation environments.

Figure 10 compares the (a) drain transient current and (b) gate transient current waveforms when an α-particle (6 MeV) strikes the S2 point (source-gate gap), with each trap type set to its critical

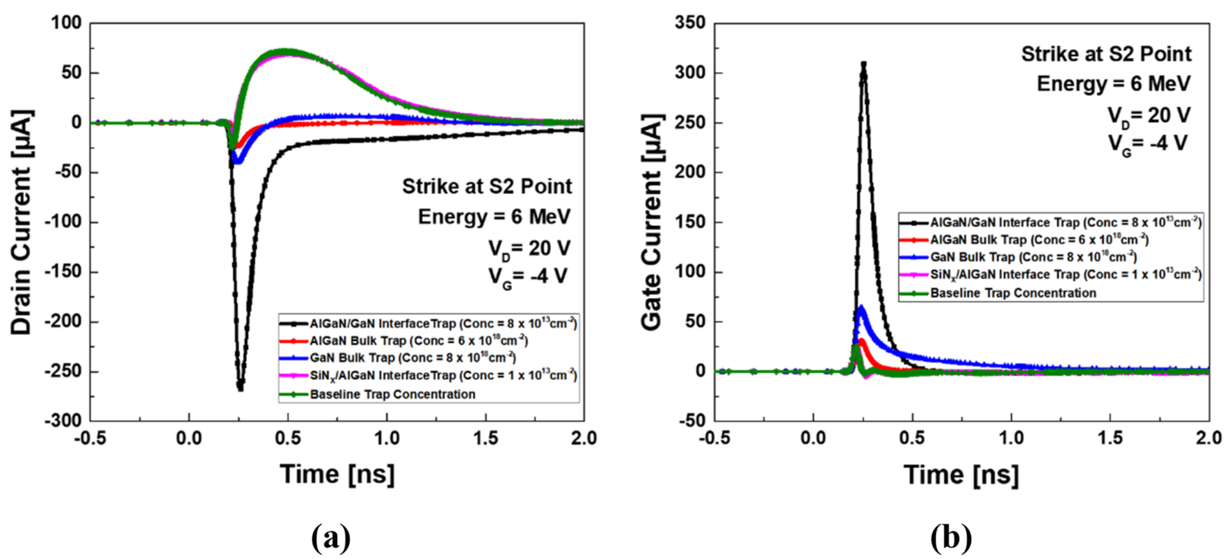

FIG. 10. Transient current responses at the S2 (source–gate spacing) location under 6 MeV α-particle incidence, with each trap set to its critical density: (a) transient drain current (I_d–time) and (b) transient gate current (I_g–time).

density value. Consistent with previous results, a distinct change in the waveform is apparent for the acceptor-type traps (AlGaN/GaN interface, AlGaN bulk, and GaN bulk). Under identical conditions, the transient associated with the surface donor trap nearly overlaps with the baseline, showing negligible deviation.

This difference arises because the strike location, S2, directly interacts with the energy barrier and the interfacial electric field beneath the gate electrode. The acceptor-type traps effectively capture the electrons generated by the α-particle strike, forming a negative space charge. This, in turn, redistributes the electric field distribution inside the device and alters the transport path of the electrons. In particular, the gate transient current observed in Fig. 10(b) suggests that a portion of the electrons, which should have originally been collected by the drain, were diverted into a gate leakage path.

The gate leakage current becomes most significant when the density of the AlGaN/GaN interface trap exceeds a critical value (8×10^{13} cm^{-2}). Figure 11 provides the physical basis that supports this change in electrical characteristics. Figure 11(a) shows the electric field profile observed at the critical density for each of the three

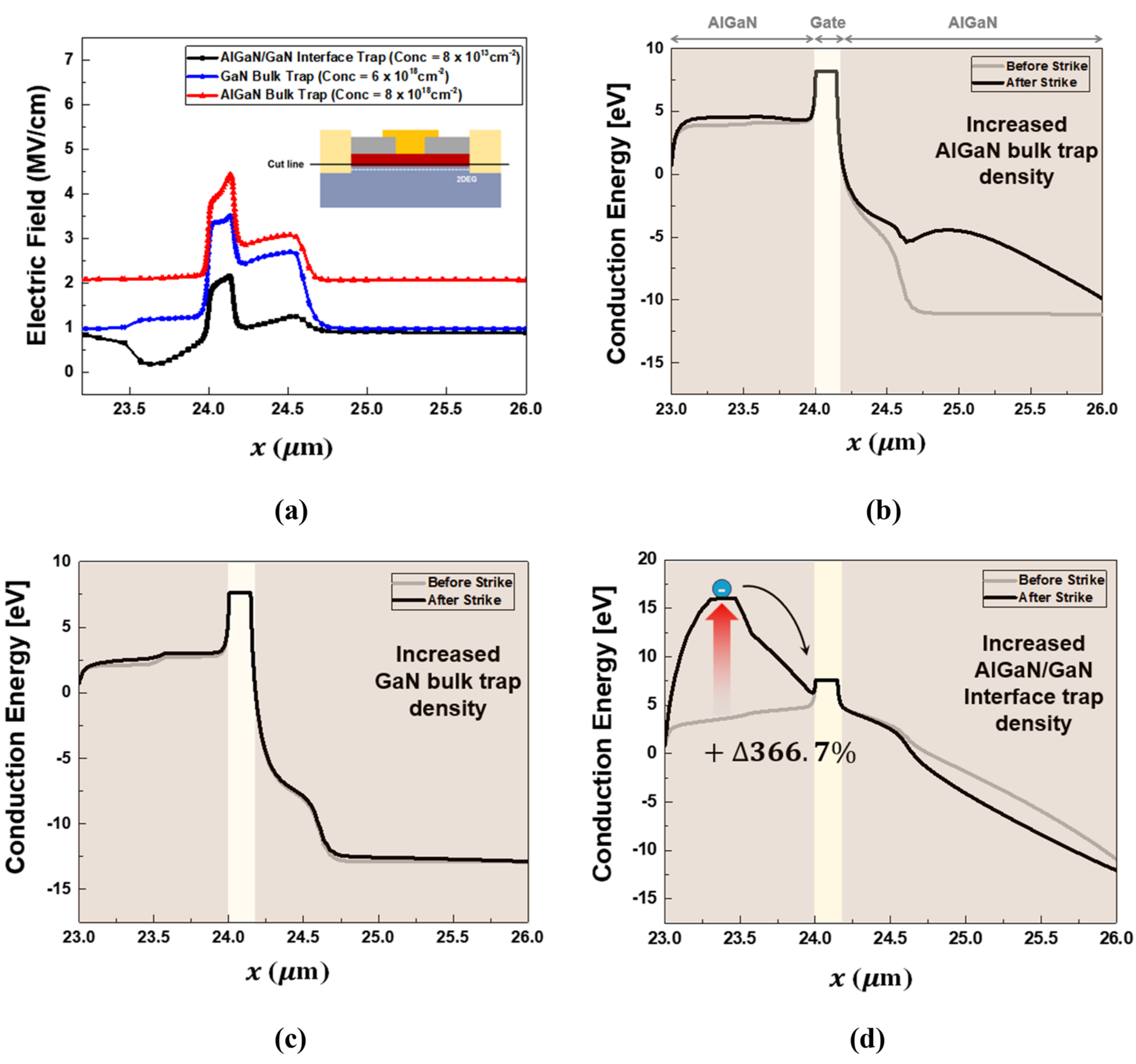

FIG. 11. Comparison of electric field distribution and conduction band profile variations at the S2 (source–gate spacing) location before and after α-particle incidence, for different trap types: (a) Electric field distributions for each trap type at their critical density before incidence. (b) Conduction band profile variations before and after incidence with AlGaN bulk trap density. (c) Conduction band profile variations before and after incidence with GaN bulk trap density. (d) Conduction band profile variations before and after incidence with AlGaN/GaN interface trap density.

trap types before the particle strike. Electrons trapped at interface acceptor states partially neutralize the built-in polarization charge, redistributing the local electric field.

This electric field redistribution directly affects the conduction band energy profile. Particularly in the case of Fig. 11(d), where the interface trap is present, a severe downward band bending phenomenon occurs in the conduction energy band beneath the gate electrode. This effectively reduces both the height and width of the Schottky barrier, leading to a leakage path being activated, allowing electrons to move directly to the gate electrode through the weakened barrier, which in turn induces the gate transient current.

In conclusion, the radiation transient response at the S2 location is governed by the density of the acceptor-like interface traps. The electric field redistribution caused by the trapped electrons and the subsequent intensification of band bending act as the key mechanism that amplifies the gate leakage current. Therefore, controlling and minimizing acceptor-type trap densities are essential to ensure the radiation reliability of GaN HEMTs.

IV. CONCLUSION

To the best of our knowledge, this work provides the first detailed simulation analysis of α-particle-induced SET in AlGaN/GaN HEMTs. By utilizing coupled Monte Carlo and TCAD simulations, this study advances the understanding of radiation behavior in AlGaN/GaN HEMTs. In particular, it elucidates α-particle-induced gate leakage current mechanisms and detailed sensitivities by trap type areas that have been previously unexplored in existing literature.

The simulation results revealed that the device's SET sensitivity is highly dependent on bias conditions. In the Off-state, an increased collected charge was observed compared to the On-state due to the superposition of channel depletion and the back-channel effect. Increasing drain voltage intensified the electric field at the gate–drain edge, enhancing drift velocity and thus the transient peak.

Analysis of the strike location confirmed that the transient current "hot spots" for the drain and the gate are different. The drain transient current showed its maximum value at the gate–drain edge (S4), where the electric field is concentrated. Conversely, the gate transient current was highest for a strike at the center of the gate (S3), a value that surpassed the drain peak under the same conditions. This is attributed to the localized collapse of the Schottky barrier and the creation of a defect-induced TAT path caused by the particle strike.

Notably, the analysis of trap effects clearly revealed that acceptor-like defects dominate the SET behavior. Above a critical concentration, acceptor traps at the AlGaN/GaN interface and in the bulk suppressed the drain transient current for strikes adjacent to the gate (S2, S3) by trapping electrons and depleting the 2DEG. Furthermore, trapped charges redistributed the field and distorted the energy bands, activating a leakage path and amplifying gate transients. In contrast, SiN_x/AlGaN interface traps had a negligible effect on the SET response on the nanosecond (ns) timescale.

Therefore, field-plate design and precise control of acceptor-type trap densities near the gate region are critical for radiation-hardened GaN HEMT design specifically targeting these high-alpha-flux environments.

ACKNOWLEDGMENTS

This research was supported by the National Research Council of Science and Technology (NST) grant by the Korea government (MSIT) (Grant No. GTL25051-000) and partly by Samsung Electronics Co., Ltd.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Soomin Kim and Hanggyo Jung contributed equally to this work.

Soomin Kim: Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Hanggyo Jung**: Software (equal). **Jong-Min Lee**: Resources (equal). **Jong Yul Park**: Resources (equal). **Junhyung Jeong**: Resources (equal). **Dong-Min Kang**: Resources (equal). **Jongwook Jeon**: Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

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